QUANTUM MECHANICS FROM THE JOBSON CELL MEDIUM

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FRAMING NOTE (added 2026-08-27, v4; corrected 2026-09-02):

```
The photon descriptions in this document are quantitatively correct.
The correct qualitative picture, developed in subsequent work, is that
a photon is a localized T_1g wave packet (corpuscle, Newton's term) --
not an extended sinusoidal wave. What propagates as "light" is the T_1g
TRANSVERSE vector wave in the medium (not a compression/pressure wave --
see doc_torsion Section 3.1 and doc_jobson_cell Section 7.1: the v_p
compression branch is carried by A_g, a separate scalar mode); the photon
corpuscle is the quantized, localized form of that transverse excitation.
Wave-particle duality in standard QM arises from conflating the medium
wave with the localized packet.
Note: T_1g waves may also be generated by macroscopic charge oscillations
(classical EM) without individual corpuscle events -- whether all T_1g
propagation requires a corpuscle source is not yet derived. [OPEN]
An electron is a corpuscle locked into a stable (1,2) Hopf winding;
the winding gives structure to its medium wave. A photon corpuscle carries
no persistent winding (theta=0). Both are corpuscles; the winding is the
only difference. All mathematical results in this document are unchanged.
The quantized, corpuscular nature of light -- established by the photoelectric
effect -- is consistent with localized T_1g wave packets. Newton lacked the
medium to explain the accompanying wave behavior.
```


ABSTRACT

The Schrödinger equation is not a postulate. It is the non-relativistic limit of the Klein-Gordon equation that governs Hopf winding modes in the Jobson cell medium. The derivation chain has zero free parameters:

```
1. Jobson medium: K = 1/eps_0, rho = mu_0, c = sqrt(K/rho)  [PROVEN]
2. Free pressure wave: omega = c*k  (massless) [QM1]
3. Hopf winding rest energy = m_p*c^2 -> Compton frequency omega_C = m_p*c^2/hbar
   [QM2; the Compton wavelength IS lambda_p = hbar_c/m_p = Zone 1 boundary]
4. Winding dispersion: omega^2 = c^2*k^2 + omega_C^2  [Klein-Gordon, QM3]
5. Non-relativistic limit (v << c):
     omega_KG -> omega_C + hbar*k^2/(2*m_p)
   Remove rest-mass oscillation phi = psi * exp(-i*m_p*c^2*t/hbar):
     i*hbar * dpsi/dt = -(hbar^2/2*m_p) * nabla^2 * psi  [SCHRODINGER, QM5]
6. Born rule: P(x) = |psi(x)|^2 from medium wave energy density  [QM6]
```

The Schrödinger equation is quantum mechanics. It is now derived from the same medium (Jobson cell lattice) that gives alpha, G, the Higgs mass, and quantum entanglement. No additional postulates or free parameters.

NEW GROUND: no existing interpretation of quantum mechanics -- Copenhagen, many-worlds, de Broglie-Bohm, GRW, QBism, consistent histories -- derives the Schrödinger equation, the Born rule, or the Tsirelson bound from an underlying physical medium with independently fixed mechanical properties. Every existing approach postulates the Hilbert-space/probability structure as an axiom. This paper derives it instead, as a mechanical consequence of the same medium that gives alpha, G, and the Higgs mass in the companion papers.

Consequences:

```
- Double-slit: Hopf winding fields span both slits simultaneously.
  Interference = constructive/destructive Huygens superposition in medium.
- Which-path: Zone 3 field decoherence resolves winding at r < r_lock(T).
  r_lock(300K) = 340 fm; coherence preserved for macroscopic slit separations.
- Delayed choice: no paradox. Medium configuration never had a single-slit
  address. "Measurement" = local perturbation resolving global winding.
  Timing is irrelevant; topology is not retroactively changed.
```

NEW PREDICTIONS (not in standard QM):

```
- Minimum electron slit width for coherent diffraction: 2*lambda_bar_e = 772 fm
  (below this, slit-edge Zone 3 fields decohere the electron winding)
- Minimum proton slit width: 2*lambda_bar_p = 0.421 fm = r_grind (nuclear hard core)
- Fringe visibility vs temperature: V(T) ~ exp(-r_sep / r_lock(T))
  where r_lock(T) = (alpha*hbar_c*r_p^2 / k_B*T)^(1/3)
```

Companion script: analysis/demos/qm_doc.py (21/21 PASS)

1. MEDIUM WAVE EQUATION AND KLEIN-GORDON

1.1 Free pressure waves

The Jobson cell medium has bulk modulus K = 1/epsilon_0 and density rho = mu_0.

The pressure wave speed c = sqrt(K/rho) = 1/sqrt(epsilon_0*mu_0) (exact by SI).

Free waves satisfy:

```
(d^2/dt^2 - c^2 * nabla^2) phi = 0
```

Dispersion: omega = c * k (massless, linear). [QM1]

(Derived from a discrete Jobson-cell lattice in the continuum limit, not an

independent postulate -- analysis/quantum/lattice_dwell_time_bridge.py.)

1.2 Hopf winding rest energy -> mass gap

A Hopf winding mode (the proton) is a topological defect in the Jobson cell

medium with winding number (1,2) (p=1, q=2). The Maxwell jamming constraint

(N_J = 21 cells, 3V-E = 6 exactly critical) sets the minimum size of the

winding. Below this size, the winding cannot be compressed -- this creates

a minimum energy:

```
E_0 = m_p * c^2  [proton rest energy = mass gap of the winding mode]
```

This generates the Compton oscillation frequency:

```
omega_C = m_p * c^2 / hbar
```

Crucially: the Compton wavelength lambda_bar = hbar / (m_p * c) = hbar_c / m_p

IS the Zone 1 boundary lambda_p. The winding resonates at the Compton frequency

WITHIN Zone 1 (the Maxwell critical jamming region). This is the same lambda_p

that appears in r_p = 4*lambda_p (PS4) and throughout nuclear structure.

The identification lambda_bar = lambda_p is not a coincidence: it follows from

the proton being the Maxwell-critical Hopf winding in the Jobson cell medium.

[QM2 PASS: omega_C = c / lambda_p confirmed to machine precision]

1.3 Klein-Gordon equation for the winding mode

With a mass gap, the dispersion relation becomes:

```
omega^2 = c^2 * k^2 + omega_C^2    [Klein-Gordon]
```

At k=0 (standing winding): omega = omega_C (pure Compton oscillation)

At k >> omega_C/c: omega -> c*k (ultra-relativistic, massless limit)

This IS the Klein-Gordon equation: (d^2/dt^2 - c^2*nabla^2)phi = -(m_p*c^2/hbar)^2*phi

[QM3 PASS: both limits verified]

(Both dispersion relations above are derived from a discrete Jobson-cell

lattice model, not postulated -- lattice_dwell_time_bridge.py BR1-BR4.)

2. NON-RELATIVISTIC LIMIT: SCHRODINGER EQUATION

For non-relativistic motion (v << c, k << omega_C / c = m_p*c / hbar):

```
omega_KG = sqrt(c^2*k^2 + omega_C^2)
         = omega_C * sqrt(1 + (hbar*k / m_p*c)^2)
         ~ omega_C + hbar*k^2 / (2*m_p) + O((v/c)^4)
```

The omega_C term = rest-mass oscillation. Extract it:

```
phi(x,t) = psi(x,t) * exp(-i * m_p*c^2*t / hbar)
```

Substituting into Klein-Gordon and dropping the O((v/c)^2) correction to d^2psi/dt^2:

```
i*hbar * d(psi)/dt = -(hbar^2 / 2*m_p) * nabla^2 * psi
```

This IS the time-dependent Schrödinger equation for a free particle. [QM5 PASS]

Accuracy: The approximation is exact to order (v/c)^2. For v/c = 1/1000 (deep NR):

relative error = 2.5e-7, consistent with the (v/c)^2/4 Taylor expansion.

ZERO additional assumptions were made. The Schrödinger equation follows from:

```
(a) The Jobson cell medium (K, rho, c -- already proven)
(b) The Hopf winding mass gap E_0 = m_p*c^2 (from Maxwell jamming -- already proven)
(c) The non-relativistic approximation (standard physics -- no extra postulate)
```

Potentials (V != 0): adding an external pressure field V(x) to the medium gives:

```
i*hbar * dpsi/dt = -(hbar^2/2m)*nabla^2*psi + V(x)*psi
```

This is the full Schrödinger equation. V(x) = the local Jobson cell pressure

(Coulomb, gravity, nuclear) -- all derived from the same medium.

3. BORN RULE FROM WAVE ENERGY DENSITY

In the Jobson medium, the energy density of a pressure wave is:

```
u(x,t) = (1/2)*rho*v^2 + (1/2)*K*strain^2 proportional to |phi(x,t)|^2
```

The probability of a detector at x absorbing the wave (detecting the particle)

is proportional to the wave energy density at x:

```
P(x) = u(x) / integral u dV = |psi(x)|^2 / integral |psi|^2 dV  [Born rule]
```

This is the Born rule. It requires no additional postulate: detection = energy

absorption from the medium wave, and energy density is proportional to amplitude squared.

[QM6 PASS: Born rule from medium energy density]

Note on wave identity: Light propagation = T_1g transverse vector wave in the Jobson cell medium (Klein-Gordon with m=0 -> Maxwell equations; see doc_torsion Section 3.1 -- T_1g is transverse, not the v_p compression branch, which is carried by A_g). A photon is a localized T_1g wave packet (corpuscle) that IS one source of such waves;
macroscopic charge oscillations (classical EM) may generate T_1g waves through the cog/Zone 3 mechanism without individual corpuscle emission events. Whether all T_1g waves require a corpuscle source or can be generated purely classically is not yet derived from the framework. [OPEN: classical vs quantum T_1g generation -- Series 3 target.] Matter (electrons, protons) = massive Hopf winding modes (Klein-Gordon with mass gap m*c^2). Light waves and matter windings are two excitation types of the same medium, not separate entities.
The photon (corpuscle) is the theta=0 mode. Here theta is the winding tilt angle: it measures how far the Hopf winding departs from the massless T_1g baseline. theta=0 means no Zone 1 displacement and no rest mass; theta>0 means the winding displaces Zone 1 and acquires rest energy m*c^2.
Formula: theta(m) = arcsin(8*alpha*phi*m/m_p). All matter requires theta > 0.

Pair production: when a photon's pressure wave amplitude exceeds the threshold for nucleating a new Hopf winding (E >= 2*m_e*c^2 for e+e- pair), the medium forces a new topological winding into existence. This is the Schwinger mechanism: field amplitude > winding nucleation threshold.
Environments where this occurs: gamma-ray burst cores (kT ~ MeV), neutron star merger ejecta, the early universe (T > 5e9 K ~ 0.4 MeV), and extreme laser fields approaching the Schwinger critical field (E_S = m_e^2*c^3/e*hbar).
Note: the solar core (kT ~ 1.3 keV) is ~800x below threshold and does NOT continuously pair-produce. [See doc_nucleus Zone 2: N_J=21 for the winding nucleation condition; open_items.txt F-12 for the photon-trapping calculation]

4. DOUBLE-SLIT EXPERIMENT

4.1 Why interference occurs

The proton (or electron) Hopf winding mode in the medium has a Zone 3 field extending to infinity (falling as 1/r^2). When the winding approaches a barrier with two slits:

```
(a) The winding CANNOT be "at slit A" or "at slit B" -- it is a topological
    configuration of the medium, defined globally (same as the A_g singlet
    in doc_entanglement).
(b) The medium pressure field of the winding passes through BOTH slit openings
    simultaneously.
(c) At the screen, the pressure wave contributions from both slits superpose:
    constructive where path difference = n*lambda_dB, destructive otherwise.
(d) Interference pattern = standard Huygens-Fresnel diffraction for the medium
```

```
pressure wave. Fringe spacing = lambda_dB / d where d is slit separation.
lambda_dB = 2*pi*hbar / (m*v) = de Broglie wavelength (from KG dispersion).
```

4.2 Why which-path measurement destroys interference

From doc_entanglement EP3/EP6: a detector at one slit places a Zone 3-active particle (detector atoms) within the phase-locking radius r_lock(T) of the beam. The Zone 3 field coupling resolves the global winding into a definite path.

```
r_lock(300K) = 340 fm  [Phase-lock threshold at room temperature]
```

For macroscopic slits (separation >> 340 fm), the detector itself must physically interact with the beam to resolve the path. Simply placing a detector nearby does NOT decohere (consistent with observation: detection requires interaction).

4.2a "Sneakily turning off" the detector does not restore interference

A crucial implication: decoherence is caused by PHYSICAL ZONE 3 COUPLING between the particle's winding and detector atoms, NOT by whether a human reads the output.

```
Experiment: a which-path detector is "turned off" (recording stopped) but the
detector atoms remain physically in place near the beam path. The particle
pattern (two bands, no interference) persists.

Explanation:
  Detector ON and recording:   atoms within r_lock -> winding resolved -> particle pattern
  Detector "off" (not recording) but atoms present: atoms STILL within r_lock
                                -> Zone 3 coupling STILL occurs -> winding STILL resolved
                                -> particle pattern persists. "Off" means nothing.
  Detector physically removed: no atoms within r_lock -> no coupling -> interference returns.

The particle's winding does not "know" whether a human is reading the detector.
It only responds to whether another Zone 3 field is within 340 fm.
The threshold is r_lock(T) -- a physical scale, not an epistemic one.

This resolves a longstanding confusion in quantum measurement theory:
measurement = physical interaction at r < r_lock(T), not observation.
Copenhagen is vague on this distinction. Torsionverse is not.

ALSO EXPLAINS: the quantum eraser (Kim et al. 2000). Pairs whose idler photon
went through the eraser (no Zone 3 coupling forced) retain the A_g singlet ->
their sub-ensemble shows interference in coincidence counts. The "recovery" of
the pattern is not retroactive: those pairs were never decohered to begin with.
```

4.3 New prediction: minimum slit width

Below 2*lambda_bar (reduced Compton wavelength), the slit-edge Zone 3 fields of slit atoms become comparable to the winding's Zone 3 field -> decoherence.

```
Electron: 2*lambda_bar_e = 2*hbar_c/m_e = 772 fm  [QM7]
Proton:   2*lambda_bar_p = 2*hbar_c/m_p = 0.421 fm = r_grind  [QM8]
```

Standard QM has no minimum slit width. Torsionverse predicts one, testable with sub-pm electron diffraction experiments.

5. DELAYED CHOICE AND QUANTUM ERASER

5.1 Delayed choice: paradox dissolved

Wheeler's delayed choice: after the particle passes the slits, the experimenter decides whether to measure which-path or allow interference.

In the torsionverse:

```
- The "particle" is a Hopf winding mode in the medium
- After passing the slits, the medium configuration spans BOTH slit openings
- There is no "particle at slit A or B" -- only a global medium configuration
- The choice of measurement = whether to apply a Zone 3-resolving perturbation
```

The timing of the perturbation is irrelevant:

```
- "Before": resolve now -> no interference (winding localized)
- "After": resolve after slits -> medium was never in a single-slit state
  The resolution perturbs the CURRENT medium state, not a past state
- No retrocausality: the medium is Markovian (current state determines future)
```

[QM9 PASS: no retrocausality required]

5.2 Quantum eraser

When which-path information is "erased" (the resolving perturbation is undone before it propagates), the A_g global winding was never permanently resolved. The winding continues in its global configuration. Interference returns.

This is not mysterious: the perturbation must propagate classically at c to affect the screen. If it is undone first, it never reaches the screen.
[Consistent with quantum eraser experiments by Kim et al. 2000]

6. STATUS SUMMARY

PROVEN (from existing medium derivations + this document):

```
- Schrodinger equation from Klein-Gordon NR limit  [QM4-5, 10/10 PASS]
- Born rule from medium wave energy density  [QM6]
- Double-slit interference mechanism (Huygens in medium)
- Delayed choice: no retrocausality  [QM9]
```

ESSENTIALLY DERIVED (borrowed from entanglement_doc):

```
- Which-path decoherence at r < r_lock(T)  [entanglement_doc EP3; essentially
  derived -- t in MeV open; QM10 check is r_lock>0 sanity only]
```

NEW PREDICTIONS (physical argument from Zone 3 geometry; not yet experimentally tested):

```
- Minimum electron slit width 2*hbar_c/m_e = 772 fm  [QM7 -- script check
  verifies formula; labeled 'NEW PREDICTION' in script; Zone 3 argument]
- Minimum proton slit width 2*lambda_p = r_p/2 = r_grind  [QM8 -- second
  condition IS a real check: r_p = 4*lambda_p (PS4); first condition is
  formula-only]
```

ESSENTIALLY CLOSED:

```
- Full Schrodinger equation with potential V(x): V = local medium pressure
  (Coulomb, gravity all derived from same medium; adding V is direct)
- Quantum eraser: perturbation propagation bounds the erasure window
- Pauli equation: from Dirac NR limit + E+ spinor [qm_dirac_pauli.py QD6]
- Dirac equation: from Clifford (T_1g generators) + E+/E- (2I) [QD1-QD4]
  g=2 from (sigma.v)^2 = |v|^2 [QD5]; proton g_p != 2 from Zone structure
- Path integral: from medium Green's function (C7 static limit = Coulomb)
  Composition rule, classical action phase, H E_1 = -13.6 eV [QP1-QP5]
```

GENUINELY OPEN:

```
- Fringe visibility V(T) quantitative formula from r_lock(T)
- Relativistic QM with spin: Dirac + EM = QED (next paper)
```

7. COMPANION SCRIPT

```
analysis/demos/qm_doc.py    -- 21/21 PASS  [STANDALONE -- no external deps]

Working scripts:
analysis/quantum/qm_from_medium.py    -- 10/10 PASS  [KG -> Schrodinger, Born rule]
analysis/quantum/qm_dirac_pauli.py    --  7/7  PASS  [Clifford, Dirac, Pauli, g=2]
analysis/quantum/qm_path_integral.py  --  5/5  PASS  [propagator, composition, E_1]
```

REFERENCES


```
[doc_qm]            https://doi.org/10.5281/zenodo.22052395
[doc_entanglement]  https://doi.org/10.5281/zenodo.22052350
[doc_torsion]       https://doi.org/10.5281/zenodo.22016573
[doc_alpha]         https://doi.org/10.5281/zenodo.22013651
[doc_nucleus]       https://doi.org/10.5281/zenodo.22042337
```


Repository: https://doi.org/10.5281/zenodo.22105622

Code: https://github.com/Destraag/torsionverse_public